WEEK 7 REPORT

STUDENT PERFORMANCE-DATA CLEANING AND TRANSFORMATION

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1. INTRODUCTION

Data cleaning and transformation are important steps in any data analysis process. Before creating charts, dashboards or statistical calculations, the data should be checked and prepared properly. Incorrect data types, duplicate records, missing values and inconsistent categories can affect the accuracy of the final analysis.

For this Week 7 activity, the Student Performance in Exams dataset was imported into Microsoft Power BI Desktop and processed using Power Query Editor. The dataset contains information about students along with their Mathematics, Reading, and Writing scores.

The main purpose of this task was to clean the dataset, transform the available columns, create new calculated columns, and prepare the data for analysis. In addition, DAX measures were created to calculate subject-wise averages, overall average performance, and the overall pass percentage.

The completed data was finally loaded into the Power BI report and displayed using KPI cards and a subject comparison chart. Slicers were also used to make the report interactive.

2. OBJECTIVES

The main objectives of this Week 7 task are:

1. To import the Student Performance dataset into Power BI Desktop.
2. To inspect the dataset and understand the different columns and their data types.
3. To identify and remove duplicate records.
4. To check for missing and invalid values.
5. To correct the data types of the score columns.
6. To standardize and prepare categorical attributes such as gender, race/ethnicity, parental level of education, lunch and test preparation course.
7. To create a Total Score column by adding Mathematics, Reading and Writing scores.
8. To create an Average Score column using the Total Score.
9. To classify students into different Performance Grade categories.

10. To create a Pass/Fail classification based on the selected average score criterion.
11. To unpivot the subject score columns into Subject and Score columns.
12. To create DAX measures for Mathematics, Reading, Writing, Overall Average and Overall Pass Percentage.
13. To validate the results using Power BI visuals.

3. DATASET DESCRIPTION

The dataset used for this activity is the Student Performance in Exams dataset. It contains student-related information and their examination scores.

The main categorical attributes used in the dataset are:

• Gender • Race/Ethnicity • Parental Level of Education • Lunch • Test Preparation Course

The numerical score attributes are:

• Math Score • Reading Score • Writing Score

These three subject scores were used to calculate the Total Score and Average Score of each student.

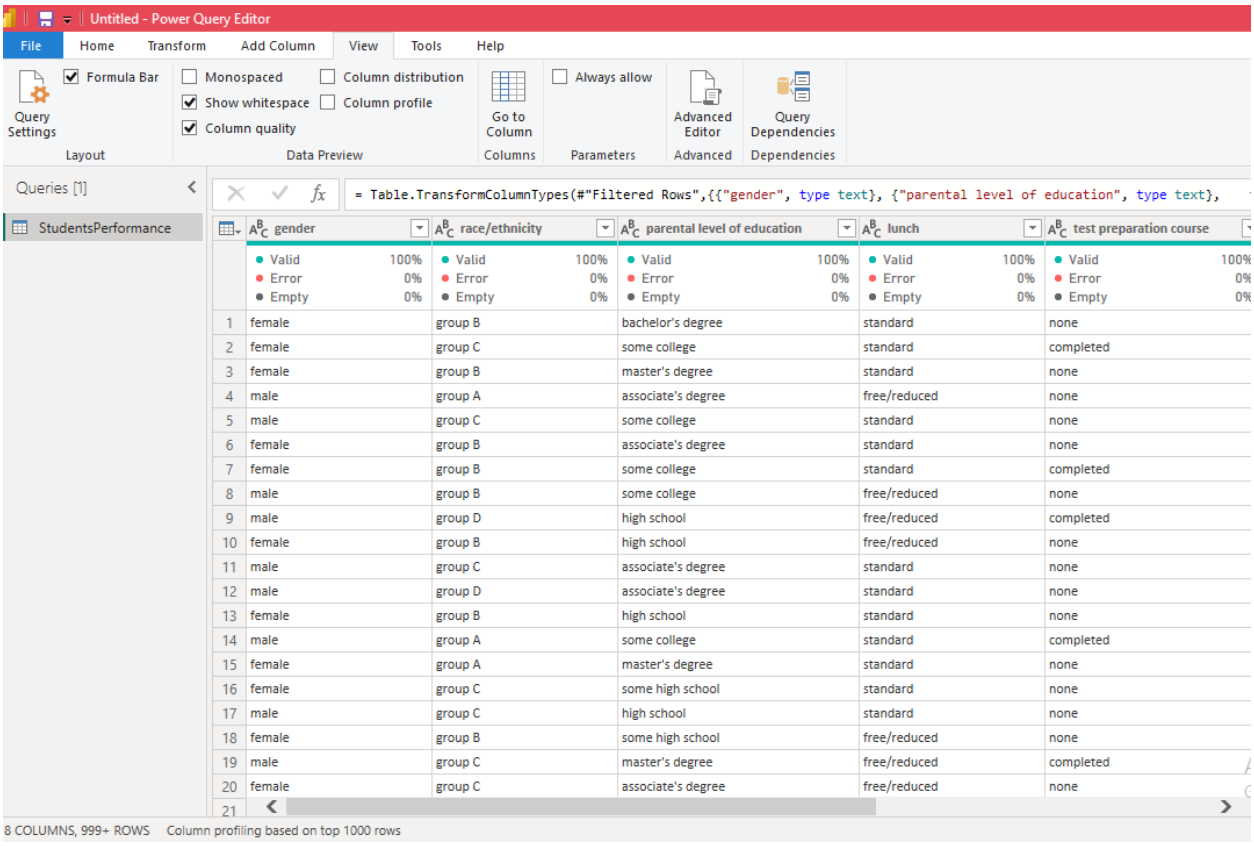
The dataset was processed in Power Query Editor before being used for the final Power BI report.

4. IMPORTING THE DATA INTO POWER BI

The Student Performance dataset was imported into the Power BI Desktop. After importing the data, the Transform Data option was used to open Power Query Editor.

Power Query Editor was used because it provides tools for checking data quality, changing data types, removing duplicates, creating calculated columns and reshaping the dataset.

After opening the dataset, the columns and their values were inspected to understand the structure of the data before performing transformations.



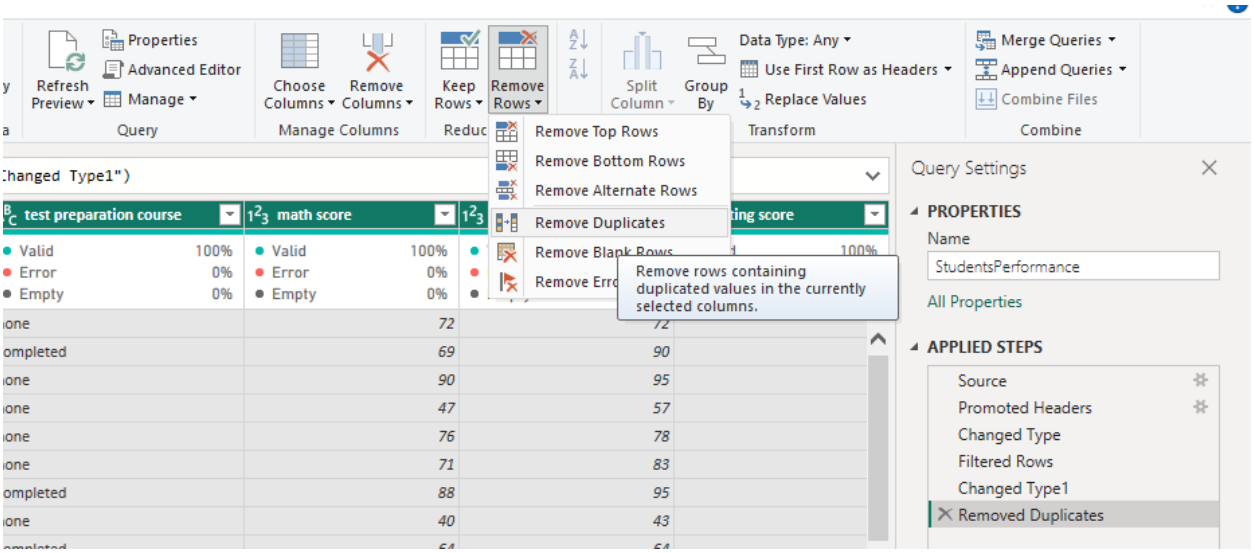
5. DATA INSPECTION AND CLEANING

The first step in the cleaning process was to inspect the columns and check the quality of the data.

The categorical columns were checked for valid values and empty or error values. The score columns were also inspected because they would later be used for mathematical calculations.

Duplicate records were removed using the Remove Duplicates option available in Power Query Editor. Removing duplicate records helps avoid counting the same student record more than once during analysis.

The Applied Steps section in Power Query was also used to keep track of the transformations performed on the dataset.



6. DATA TYPES AND SCORE COLUMNS

The score columns are important because they are used for arithmetic calculations. Therefore, Math Score, Reading Score and Writing Score need to be treated as numerical values.

During the transformation process, the score columns were checked and changed to appropriate numeric data types.

This step was necessary before creating the Total Score column. If the score values are stored as text, Power Query cannot directly perform addition using the + operator.

Correcting the data type ensures that the three subject scores could be added correctly.

7. CREATING TOTAL SCORE

A new calculated column named Total Score was created using Power Query's Custom Column option.

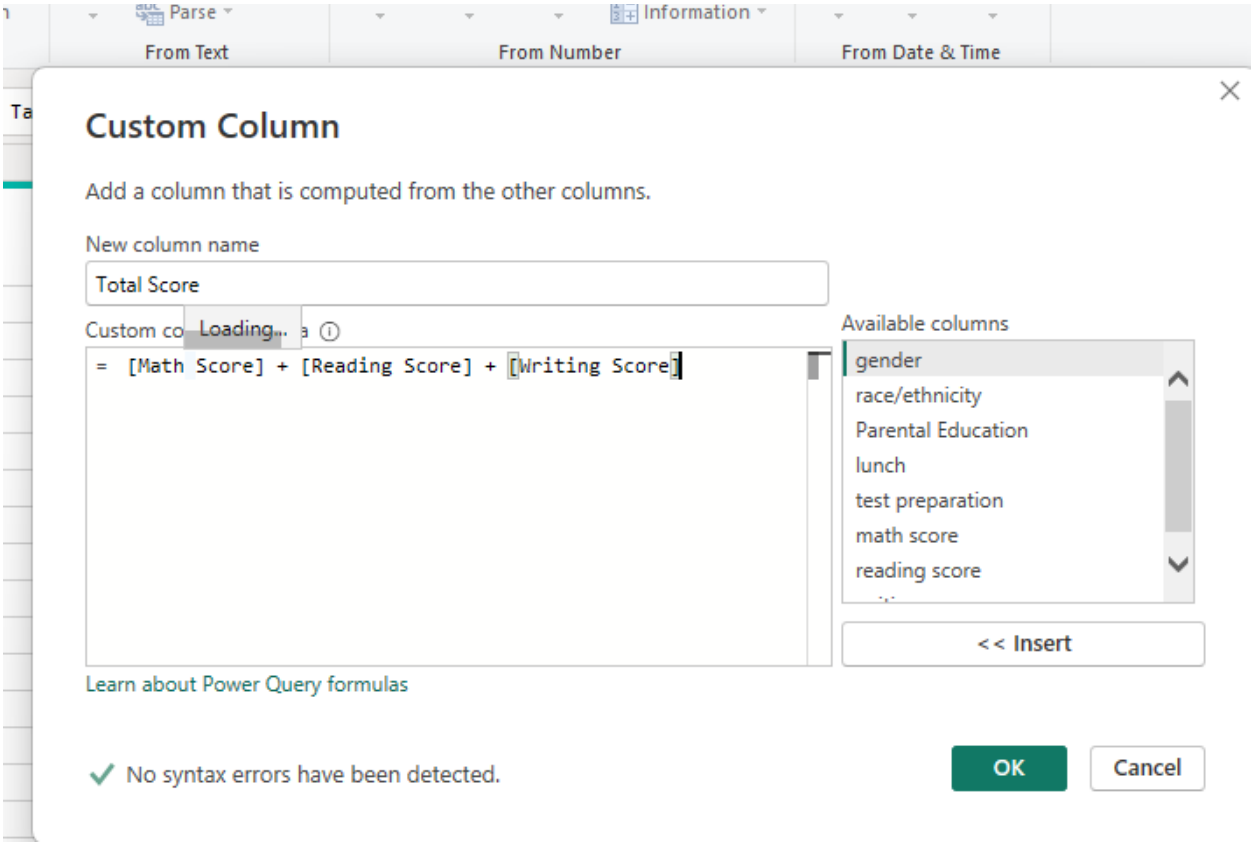
The Total Score represents the combined marks obtained by a student in Mathematics, Reading and Writing.

The formula used was:

[Math Score] + [Reading Score] + [Writing Score]

Since there are three subjects and each subject has a maximum score of 100, the maximum possible Total Score is 300.

The Total Score provides an overall view of the marks obtained by each student before calculating the average.



8. CREATING AVERAGE SCORE

After creating Total Score, another calculated column named Average Score was created.

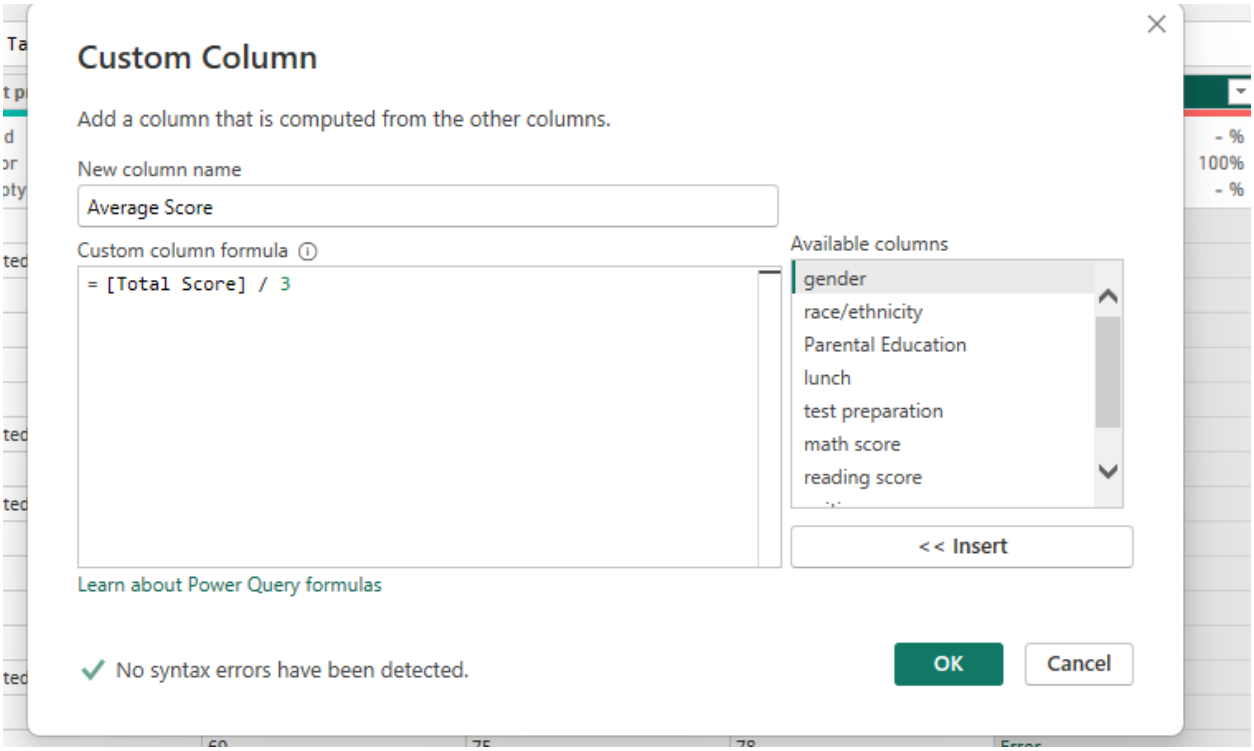
The Average Score was calculated by dividing the Total Score by 3 because the dataset contains three examination subjects.

Formula used:

[Total Score] / 3

The Average Score gives a single value representing the student's overall performance across Mathematics, Reading and Writing.

This column was later used for creating the Performance Grade and Pass/Fail classification.



9. PERFORMANCE GRADE

A Performance Grade column was created using conditional logic based on Average Score.

The grading criteria used in this implementation were:

• A – Average Score of 80 or above • B – Average Score from 60 to below 80 • C – Average Score from 40 to below 60 • F – Average Score below 40

The purpose of this classification is to make the numerical average easier to interpret.

The Performance Grade provides a simple way of grouping students according to their overall academic performance.

10. PASS/FAIL CLASSIFICATION

A Pass/Fail column was also created using conditional logic.

For this report, an Average Score of 40 or above was considered a Pass. Students with an Average Score below 40 were classified as Fail.

The formula used was:

if [Average Score] >= 40 then "Pass" else "Fail"

This classification makes it easier to calculate the overall pass percentage and analyse the number of students who meet the selected passing criteria.

12. APPLIED STEPS IN POWER QUERY

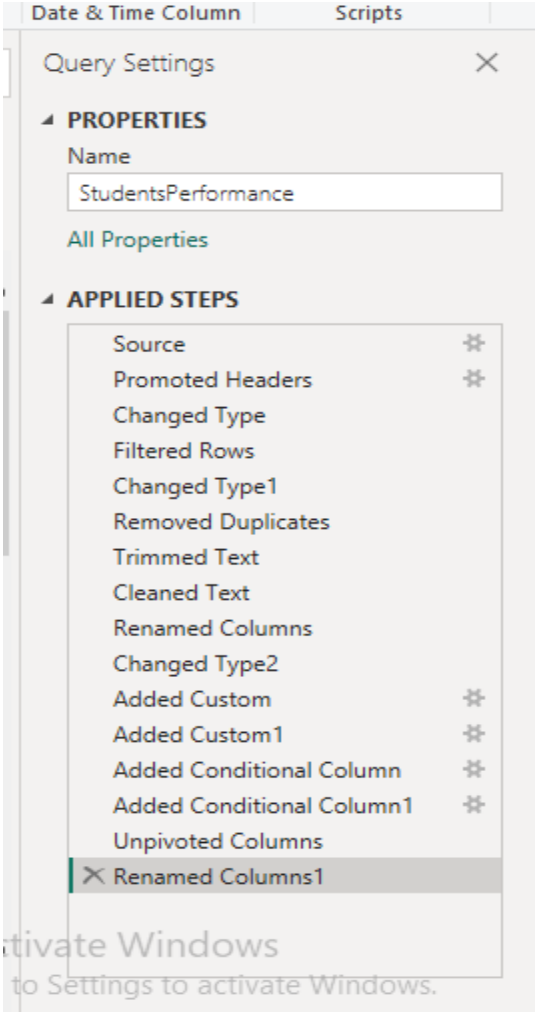
Power Query maintains an Applied Steps section that records the transformations performed on the dataset.

The transformation process included steps such as:

• Source • Promoted Headers • Changed Type • Filtered Rows • Removed Duplicates • Trimmed Text • Cleaned Text • Renamed Columns • Changed Data Types • Added Custom Columns • Unpivoted Columns • Renamed Columns

The Applied Steps section is useful because it provides a record of how the original dataset was transformed into the final analysis-ready table.

This also makes the transformation process easier to review and reproduce.



13. UNPIVOTING THE SUBJECT SCORES

The original dataset contained separate columns for Math Score, Reading Score and Writing Score.

To make the data more suitable for analysis and visualization, these subject columns were unpivoted.

After unpivoting, the score information was represented using two columns:

• Subject • Score

The Attribute column was renamed to Subject and the Value column was renamed to Score.

This transformation changes the structure of the data from having separate columns for each subject to having one common Score column and one Subject column.

This structure is useful when creating charts that compare different subjects because the same Score field can be used for all three subjects.

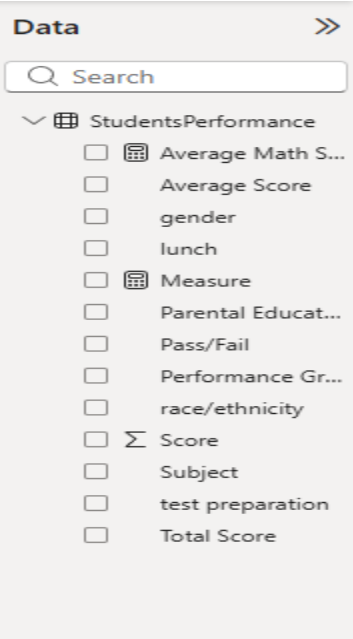
14. FINAL DATA TABLE

After completing the transformations, the final table was available in Power BI under the name StudentsPerformance.

The table contains the categorical attributes, Subject, Score, Total Score, Average Score, Performance Grade and Pass/Fail.

The DAX measures created for the report are also stored under the StudentsPerformance table.

The final data structure was checked before creating the report visuals.



15. DAX MEASURES

After completing the data transformation, DAX measures were created in Power BI to calculate important performance indicators.

Five measures were created:

1. Average Math Score
2. Average Reading Score
3. Average Writing Score
4. Overall Average Score
5. Overall Pass %

Since the subject scores were unpivoted into Subject and Score, the first three measures calculate the average Score after filtering the Subject field.

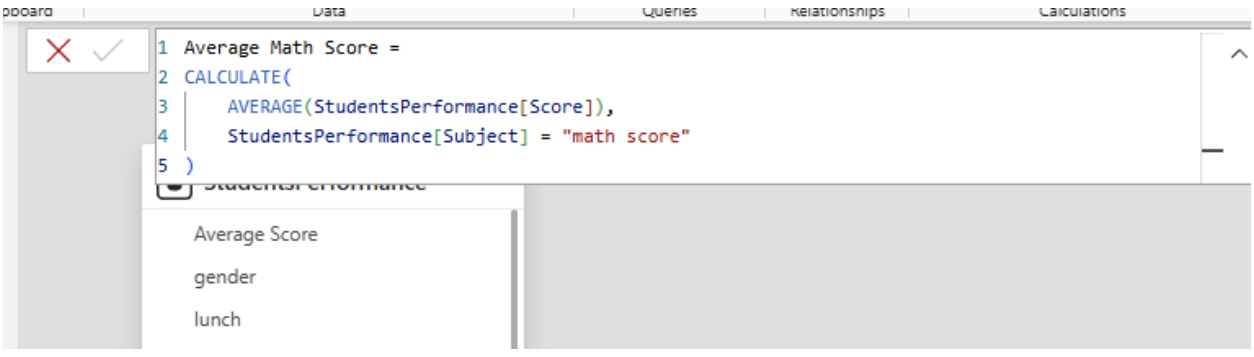
16. AVERAGE MATH SCORE

The Average Math Score measure calculates the average Mathematics score from the Score column while filtering the Subject column for Mathematics.

DAX formula:

Average Math Score = CALCULATE(AVERAGE(StudentsPerformance[Score]), StudentsPerformance[Subject] = "math score")

This measure was used in a KPI card to display the overall average Mathematics performance.



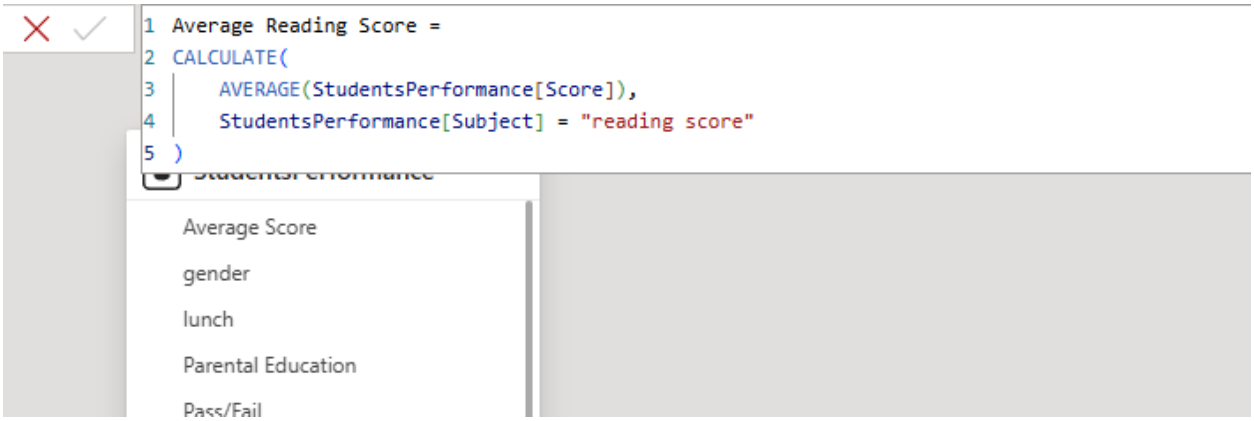
17. AVERAGE READING SCORE

The Average Reading Score measure calculates the average Reading score.

DAX formula:

Average Reading Score = CALCULATE(AVERAGE(StudentsPerformance[Score]), StudentsPerformance[Subject] = "reading score")

This measure allows the Reading performance to be displayed separately in the Power BI report.



18. AVERAGE WRITING SCORE

The Average Writing Score measure calculates the average Writing score using the Score column and a filter on the Subject field.

DAX formula:

Average Writing Score = CALCULATE(AVERAGE(StudentsPerformance[Score]), StudentsPerformance[Subject] = "writing score")

This measure was displayed as one of the KPI cards in the final report.



19. OVERALL AVERAGE SCORE

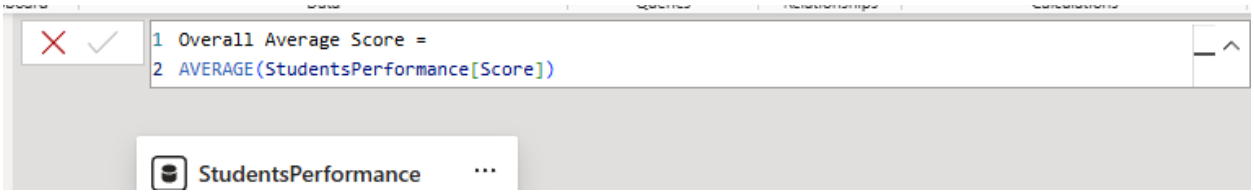
The Overall Average Score measure calculates the average value of the Score column.

DAX formula:

Overall Average Score = AVERAGE(StudentsPerformance[Score])

This provides an overall average across the subject scores available in the unpivoted table.

The result was displayed using a KPI card in the Power BI report.



20. OVERALL PASS PERCENTAGE

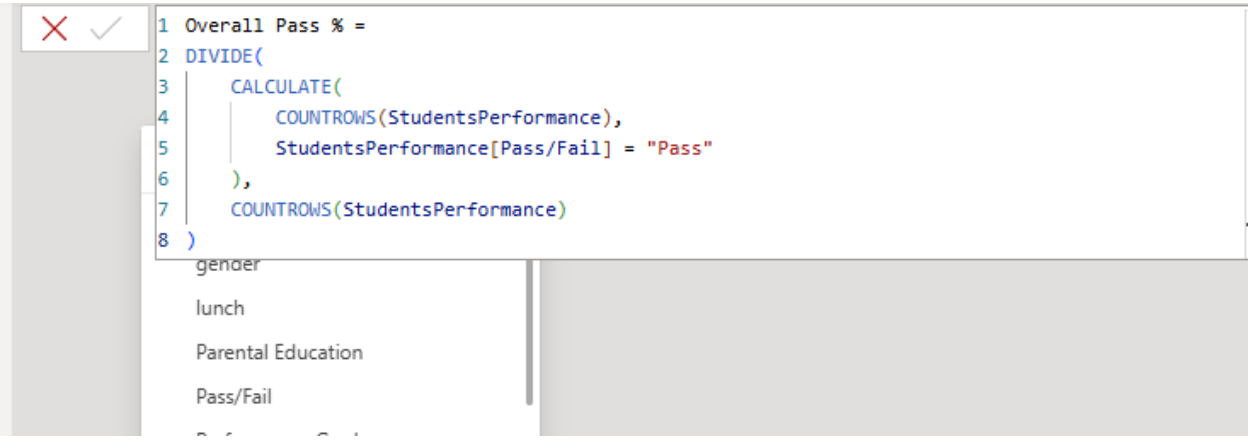
The Overall Pass % measure calculates the proportion of records classified as Pass.

DAX formula:

Overall Pass % = DIVIDE(CALCULATE(COUNTROWS(StudentsPerformance), StudentsPerformance[Pass/Fail] = "Pass"), COUNTROWS(StudentsPerformance))

The measure uses DIVIDE to calculate the number of Pass records divided by the total number of records.

The result can be formatted as a percentage in Power BI.



21. FINAL POWER BI REPORT

After completing the data transformation and creating the DAX measures, the calculated values were added to the Power BI report using Card visuals.

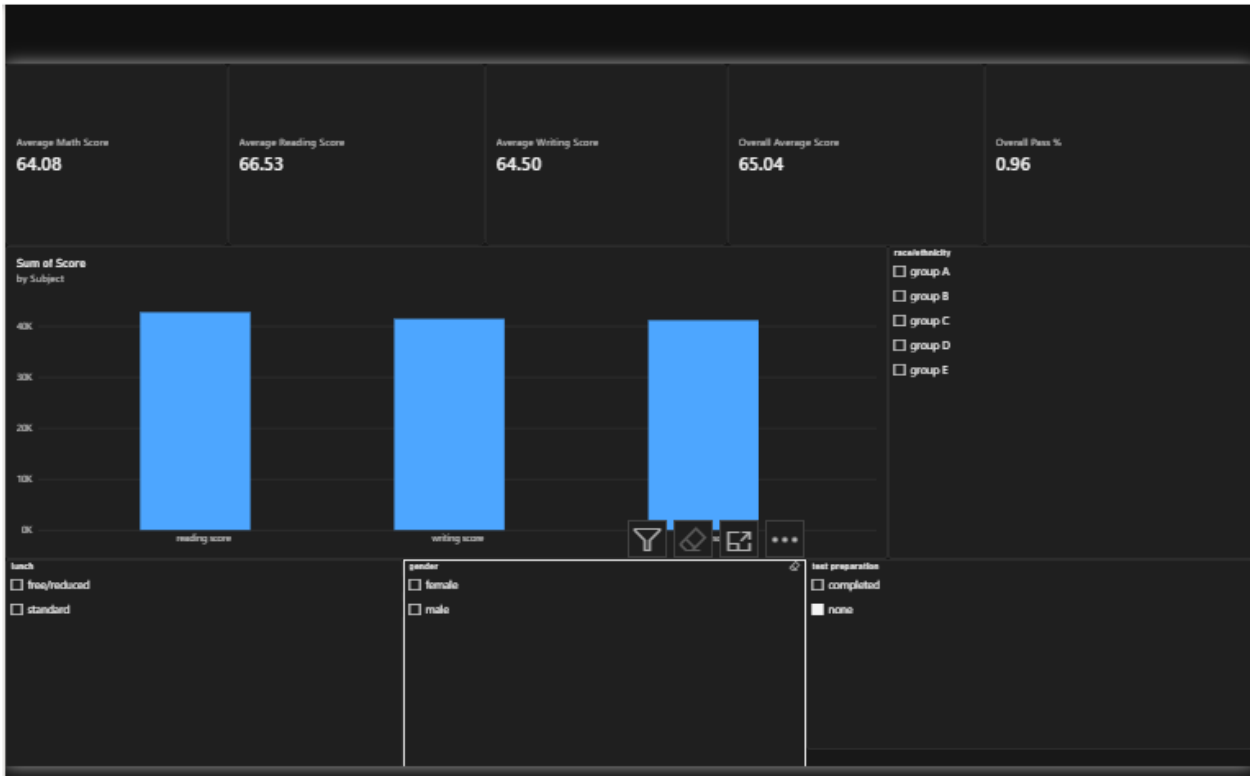
The final report contains KPI cards for:

• Average Math Score • Average Reading Score • Average Writing Score • Overall Average Score • Overall Pass %

A subject comparison chart was also included to compare the total scores across Reading, Writing and Mathematics.

Slicers were added for different categorical attributes including Race/Ethnicity, Lunch, Gender and Test Preparation Course. These slicers allow the user to filter the report and examine the performance of different groups.

The final report provides a simple interactive view of the student performance dataset.



27. CONCLUSION

The Week 7 Student Performance data cleaning and transformation task was successfully completed using Microsoft Power BI Desktop.

The dataset was first inspected and cleaned in Power Query. Duplicate records were removed and the score columns were converted into appropriate numeric data types. Calculated columns were then created for Total Score, Average Score, Performance Grade and Pass/Fail.

The Math, Reading and Writing score columns were unpivoted into Subject and Score columns. This created a more suitable structure for analysis and visualization.

Five DAX measures were created to calculate the average Mathematics score, average Reading score, average Writing score, Overall Average Score and Overall Pass Percentage.

Finally, the measures were displayed using KPI cards and the subject scores were compared using a chart. Slicers were also added to make the report interactive.

The completed report provides a clear summary of student performance and demonstrates the complete process from raw data preparation to Power BI reporting.